

HARSHAL KHARABE

Python Backend Developer

harshkharabe007@gmail.com | +91 7028653774 | Pune, Maharashtra | linkedin.com/in/harshalkharabe
github.com/hkcoder18

PROFESSIONAL SUMMARY

Backend Engineer(Python) with 2+ years of production experience building scalable REST APIs, event-driven data pipelines, and workflow orchestration systems using Python and FastAPI. At DataMatter Technologies, shipped 15–20 production-grade APIs supporting multi-tenant authentication, RBAC, and async data workflows. Designed an SQS-based event-driven architecture to handle GB-scale CSV ingestion and built a full job orchestration system with cron scheduling, retry logic, and lifecycle management. Seeking a backend-focused role at a product or startup environment where scalable systems and clean API design are valued.

EXPERIENCE

DataMatter Technologies, Python Developer - Built 15–20 production REST APIs using FastAPI — covering multi-tenant auth, RBAC, job management, and data ingestion workflows across multiple environments. - Designed and implemented tenant-based authentication to support a multi-tenant architecture, improving security and user isolation across clients. - Designed chunk-based CSV ingestion pipeline processing 1–10GB files without memory overflow, using Python streaming and PostgreSQL bulk-insert strategies. - Implemented event-driven async processing with Amazon SQS, decoupling ingestion from execution and significantly improving pipeline throughput and reliability. - Built audit logging and data integrity validation across ETL pipelines, reducing silent data errors and improving traceability across all production workflows.	04/2024 – Present Remote
--	-----------------------------

EDUCATION

Bachelor of Computer Applications (BCA), Sant Gadge Baba Amravati University CGPA: 8.12	06/2022 – 05/2025
---	-------------------

TECHNICAL SKILLS

Languages: Python, SQL Backend: FastAPI, Django, Flask, REST APIs Databases: PostgreSQL, MongoDB, MySQL
Data & ETL: Pandas, NumPy, chunk-based streaming, ETL pipeline design Cloud & DevOps: AWS (EC2, Lambda, SQS), Docker
Auth & Security: JWT, RBAC, multi-tenant architecture

PROJECTS

File Migration System, FastAPI · Amazon SQS · PostgreSQL · Pandas - Built a FastAPI File Migration system processing 1–10GB CSV files via chunk-based streaming — eliminated memory overflow entirely while maintaining data integrity across large imports. - Designed an SQS-based event-driven architecture to decouple file uploads from processing jobs, enabling scalable concurrent ingestion with zero inter-job interference. - Implemented full job lifecycle tracking (queued → processing → success/failed) in PostgreSQL with partial-failure recovery, allowing pipelines to resume mid-file without reprocessing ingested rows.
Event-Driven Workflow Orchestration Engine, FastAPI · AsyncIO · PostgreSQL - Built a workflow orchestration engine from scratch supporting cron, API-trigger, webhook, and manual execution modes — with configurable retry policies, exponential backoff, and per-job execution logs. - Implemented dynamic payload merging and runtime parameter injection, enabling flexible multi-step workflow execution without hardcoded job configurations. - Async worker pool (AsyncIO) handles concurrent job queues without thread-blocking; REST APIs expose job creation, scheduling, and live status monitoring consumed by dashboards and external services.
TrendScout – Reddit Trend Monitoring Platform, FastAPI · Python · PostgreSQL · React · APIs - Built a trend monitoring platform that aggregates and analyzes Reddit discussions to identify emerging topics, sentiment, and market trends in real time. - Designed scalable backend APIs using FastAPI for data ingestion, processing, and export workflows supporting CSV and external integrations. - Implemented export functionality for analytics data, including CSV generation and Slack integration for automated reporting and collaboration. - Developed a modular architecture with secure environment management, enabling seamless integration with third-party services and future feature expansion.